Raju Kumar Sah Aerospace Engineer

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Research Profile

Aerospace engineer specializing in experimental and computational rocket propulsion (solid, hybrid and liquid), with hands-on experience in propulsion design, precision fabrication, and static-fire testing. Led the development of Nepal's first liquid rocket engine and a national-record solid rocket; conducted propellant combustion research at **IIT Bombay**. Research interests: rocket propulsion, high-speed compressible flows, experimental combustion, and data-driven CFD. Seeking a funded graduate position to advance propulsion technology through rigorous experiment and simulation.

Honors and Awards

- **First Liquid Rocket Engine of Nepal**: LUME-X1 (2026)
- **National Rocketry Record of Nepal: Project Trishul(2025)**
- **Research Support Fund:** After winning the AIChE K-12 STEM and PRAKRITI '24 Model Exhibition
- **Mabuchi Motor Award (2023):** For most compact robot design at ABU Robocon 2023, Phnom Penh Cambodia

Research Experience

Aircraft Propulsion Laboratory, IIT Bombay Sep 2025 – Nov 2025
Research Intern — Supervisor: Prof.Dr.Nagendra Kumar *Mumbai, India*

- **Slurry propellant burn rates:** Prepared HTPB-based slurry propellant (79% AP, 17% HTPB, 3% DOA, 1% charcoal) in a sigma mixer and measured burn rates in a Crawford strand burner over 20–70 bar; fitted Saint-Robert's law by log–log regression. Pressure exponent stayed near $n \approx 0.4$ for all formulations (typical of AP/HTPB); iron oxide raised the burn-rate coefficient from 1.42 to 1.75 mm/(s·bar^n) versus the no-additive baseline.
- **Gel propellant droplet combustion:** Formulated organic (Thixotrol ST/xylene/ATF) and inorganic (SiO_2/ATF/Al) gels; burned suspended droplets under high-speed imaging and analyzed them with the d^2-law. Non-aluminized organic gels burned steadily ($R^2 > 0.98$); raising silica from 5 to 7.5% increased gel viscosity, and aluminum increased burning rate, with the most stable regression at 25% Al ($R^2 = 0.90$).
- **Laser ignition of AP/HTPB with nano-aluminum:** Ran ignition tests across laser power and ambient pressure, recording piezoelectric, photodiode and pressure-transducer signals. Ignition was inconsistent at 10–20% power and reliable from 40%; piezo ignition delay fell from about 55–76 ms at 20% power to 8 ms at 100%.
- **Electrically controlled solid propellant (ECSP):** Crawford-burner tests of a lithium-perchlorate ECSP under varying voltage, pressure and temperature.
- **Hybrid rocket motor:** Supported assembly and static testing of an HTPB/N_2O motor (chamber, nozzle, feed line, sensors) comparing showerhead and swirl injectors; reviewed equilibrium c^*, T_c and I_{sp} versus O/F ratio.
- **Sieved ammonium perchlorate** into coarse, medium and fine fractions with a tap-sieve shaker.

IIT Bombay Rocket Club (Spaceport America Cup team) Sep 2025 – Nov 2025
Propellant Preparation - Night Shift *Mumbai, India*

- Cast sorbitol/KNO_3 (35%/65%) propellant grains; introduced vibrating drip-casting that eliminated trapped bubbles and gave consistent grain density.
- Used X-ray radiography to inspect grains, identifying minor central density variations and small shrinkage voids; linked casting uniformity to reduced risk of hot spots and pressure spikes.

SRB Club (Rocketry), Pulchowk Campus Jun 2024 – May 2026
Propulsion, Structure and Fabrication Head *Kathmandu, Nepal*

- Led propulsion and structures for Projects TRISHUL, Bajra and LUME-X1, from simulation-driven design to fabrication, integration and testing.
- Performed thermochemical (NASA CEA) and CFD (ANSYS Fluent) analysis to select O/F ratios and optimize injector flow, chamber thermal loading and nozzle expansion for liquid and solid engines.
- Machined pressure chambers, nozzles, injectors, feed-line fittings, bulkheads and motor mounts on lathes and mills; fabricated glass-fiber composite airframes (body, fins, nosecone) and a lightweight pressure chamber.

Chemistry Laboratory, Pulchowk Campus, Lalitpur, Nepal Jan 2024 – May 2024
Laboratory Research Assistant *Kathmandu, Nepal*

- Screened flame-retardant compounds for cellulosic fibers and identified borax as most effective by comparative combustion testing; treated cotton showed no sustained flame versus untreated controls.
- Characterized suppression mechanisms (fused borate glass layer, endothermic heat absorption, pyrolysis-gas inhibition).

Selected Projects

LUME-X1: First Liquid Rocket Engine of Nepal (Final-Year Thesis) 2025 – 2026
Propellants: Ethanol/GOX fuel — Supervisor: Dr.Sudip Bhattrai

- *Designed, analyzed, simulated and built the engine; achieved a successful static fire on the first attempt with 92% c^* efficiency.*

Mach 4.3 Single-Ramp Hypersonic Intake Design *2025*
Python, ANSYS Fluent

- *Designed a shock-on-lip intake using oblique shock theory; automated throat sizing and θ–β–M solutions in Python, achieving 0.8466 total pressure recovery–(Design File).*
- *Validated with 2D RANS simulations (k–ω SST); mass flow matched analytical value within 1.03% (25.85 kg/s).*

National Record Solid-Fuel Rocket *2025*
Propulsion, Structure and Fabrication Head

- *Reached 2.62 km altitude, establishing a Nepal national record for the solid-fuel Rocket Trishul (2025) (Video).*

ABU Robocon 2023, Nepal Team (Robotics Club) *Aug 2022 – May 2026*
Design and Fabrication Head-Award *Phnom Penh, Cambodia*

- *Designed and machined the robot's mechanical assemblies (base plates, motor mounts, shafts, bearing housings) from CAD to integration; met a late weight limit through weight-relief pocketing and 3D-printed replacements.*

Education

Institute of Engineering, Pulchowk Campus, Tribhuvan University *2022 – 2026*
B.E. in Aerospace Engineering *Lalitpur, Nepal*

- *Aggregate: 70.88%*
- *Coursework: Compressible Aerodynamics, Advanced Propulsion, Computational Fluid Dynamics, Applied Thermodynamics and Heat Transfer, Fluid Dynamics, Numerical Methods, Control Systems, Aircraft Structures, Flight Dynamics, Aircraft Preliminary Design*
- *Thesis (final-year project): LUME-X1 liquid rocket engine(see document)*

Technical Skills

- ***Programming/Analysis:*** *Python (NumPy, SciPy, Pandas, Matplotlib), MATLAB*
- ***Simulation:*** *ANSYS Fluent, OpenFOAM, NASA CEA, OpenRocket, OpenMotor, ParaView, XFLR5*
- ***CAD:*** *CATIA, Fusion 360*
- ***Fabrication:*** *Lathe, milling, welding, composite layup, 3D printing, CNC laser cutting*
- ***Experimental:*** *Static-fire testing, propellant formulation, high-speed imaging, piezoelectric/photodiode data acquisition*

References

Prof.Dr.Nagendra Kumar, Supervisor: Dr.Sudip Bhattrai, Rocket Propulsion Elements, Thermochemical Analysis, Hypersonic